

Web-Based Healthcare Hospital Management System
Final Year Project Proposal



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Abstract

Most small clinics still rely on manual appointment booking, which leads to long waiting times, crowded waiting areas, and increased stress for both patients and clinic staff [1]. This project proposes a Web-Based Healthcare Hospital Management System that allows patients to book appointments through a website instead of visiting the clinic physically and provides digital queue tokens so patients can view their real-time position in the queue. The main objective of the system is to reduce patient waiting time and improve overall clinic management by using simple scheduling rules and basic data analysis [2]. Optional AI-based features, such as waiting time estimation and basic chatbot guidance, can also be integrated to enhance user experience [3]. The proposed system is designed to be simple, cost-effective, and well suited for small to medium-sized clinic

1. Introduction

1.1 Background of the Study

Many clinics face problems in managing patient appointments system. Most clinics still follow walk-in systems where patients come without prior booking and wait for long hours [1]. This results in overcrowded waiting areas, poor time management, and dissatisfaction among patients and clinic staff. Doctors also face difficulty in managing their schedules properly.

With the growth of web technologies, it has become easier to develop online systems that solve such problems [4]. This project introduces a Web-Based Healthcare Hospital Management System that allows patients to book appointments in advance using a web application. The system helps clinics manage patient flow efficiently by organizing appointments and queues digitally. Unlike large hospital management systems, this project focuses only on appointment booking and queue management.

Healthcare delivery in outpatient clinics often relies on walk-in visits and manual registers for appointment management [1]. This approach results in long queues, unpredictable waiting times, and increased administrative burden on clinic staff. Patients frequently arrive early and wait for extended periods without clear information about their turn. Such inefficiencies negatively affect patient satisfaction and clinic productivity.

With the widespread availability of internet access and web applications, healthcare service delivery can be improved through digital appointment systems. Web-based scheduling solutions have been shown to reduce waiting times and enhance transparency in patient flow [3]. This project builds on these concepts by providing a simple, clinic-focused system that can be easily adopted without complex infrastructure or high costs.

2. Project Motivation

The motivation for developing this project comes from real-life problems observed in clinics' manual handling of appointments, such as long patient waiting times, overcrowded waiting rooms, and frustration for patients who are unaware of their queue status [1]. Patients often waste time waiting without knowing when their turn will come, while clinic staff struggle to manage patient flow efficiently.

From an academic and technical point of view, this project provides a practical opportunity to apply concepts of web development, Firebase, and software engineering

[4]. The addition of basic AI-assisted features, such as waiting time estimation, makes the project more interesting without increasing complexity [3].

3. Aim and Objectives

3.1 Aim

The main aim of this project is to design and develop a user-friendly web-based system that automates clinic appointment booking and queue management [2]. The system aims to reduce patient waiting time, minimize overcrowding in clinics, and help clinic staff manage appointments and schedules more efficiently through digital tools.

3.2 Objectives

- To design an online appointment booking system for patients
- To provide digital queue tokens for patients
- To allow patients to view their queue position in real time
- To estimate patient waiting time using simple or basic AI techniques
- To send notifications to patients about appointment updates
- To maintain complete patient history including diagnoses, prescriptions, and past visits
- To enable structured follow-up scheduling for chronic or post-treatment patients

To provide secure online payment options including Jazz Cash, Easypaisa, and bank.

4. Scope of the Project

The scope of this project is limited to managing patient appointments and queues for outpatient clinics. The system focuses on booking appointments, generating digital queue tokens, and showing real-time queue status to patients. It does not include medical diagnosis, treatment, or detailed patient medical records [4].

The system will be developed as a web-based application that can be accessed using a web browser. It will support single-clinic and multi-doctor setups. Mobile application development and integration with large hospital management systems are not included in the scope of this project. However, the system now includes patient history tracking, follow-up management, and online payment integration via jazz Cash, Easypaisa, and bank channels to offer a more comprehensive healthcare management experience

5. Proposed Methodology

The Web-Based Healthcare Hospital Management System will be developed using a modular and web-based approach to ensure simplicity, efficiency, and ease of use [4]. The system will be divided into different modules, where each module will perform a

specific function. This approach makes the system easy to manage, test, and enhance in the future. The main modules of the system are described below:

5.1 User Registration and Authentication Module

This module will allow clinic admin to create accounts for doctors. User authentication ensures that only authorized users can access system features.

- Patients have created accounts and can login as a patient
- Patient does not authenticate, but their data is temporarily stored for appointment tracking, notifications, and queue management
- Book appointment and receive a unique appointment token
- Use token to check queue status

5.2 Appointment Booking Module

Patients will be able to book appointments online by selecting an available doctor and suitable time slot. This reduces manual booking and helps organize patient schedules efficiently.

5.3 Queue Management Module

Once an appointment is booked, the system will generate a digital queue token. The queue will be updated dynamically, allowing patients to track their position in real time.

5.4 Waiting Time Estimation Module

Basic rule-based logic or simple AI-assisted techniques will be used to estimate waiting time based on the number of patients in the queue and average consultation duration [3].

5.5 Notification Module

The system will send notifications to patients regarding appointment confirmation, queue updates, and estimated waiting time through web or SMS notifications.

5.6 Admin Dashboard

Clinic staff will use an admin dashboard to manage doctor schedules, appointments, patient queues, and system settings in an organized manner.

6. Patient History Module

The Patient History Module provides a comprehensive digital record of each patient's interactions with the clinic. It serves as a centralized repository that both doctors and authorized clinic staff can access to review a patient's medical background before or during a consultation. This module enhances clinical decision-making and eliminates the need for paper-based records.

6.1 Key Features

- **Personal Profile:** Stores patient demographics — name, age, gender, contact information, blood group, and known allergies
- **Visit History:** Logs every past appointment including date, attending doctor, diagnosis, and prescribed treatment
- **Prescription Records:** Maintains a digital record of all medicines prescribed during each visit with dosage and duration
- **Lab & Test Reports:** Allows upload and retrieval of lab results, X-rays, and other diagnostic reports
- **Chronic Condition Tracking:** Flags long-term conditions (e.g., diabetes, hypertension) for quick reference by doctors
- **Doctor Notes:** Doctors can add private clinical notes for each patient visit
- **Search & Filter:** Admins and doctors can search patient history by date range, condition, or doctor name

6.2 Data Security

Patient history is stored with role-based access control. Only the assigned doctor and authorized admin can view a patient's full record. Patients can view their own history through their personal dashboard.

Table 1: Patient History

Field	Description	Access Level
Patient ID	Unique system identifier	Admin / Doctor
Visit Date	Date of consultation	All
Diagnosis	Doctor's diagnosis notes	Doctor / Admin
Prescription	Medicines and dosage	Doctor / Patient
Lab Reports	Attached diagnostic files	Doctor / Admin / Patient
Follow-up Flag	Indicates if follow-up is needed	Doctor / Admin

7. Patient Follow-Up Module

The Follow-Up Module ensures continuity of care by allowing doctors to schedule, track, and manage follow-up appointments for patients who require monitoring after initial treatment. This is especially important for chronic disease patients, post-surgery cases, and patients on long-term medication.

7.1 Key Features

- **Follow-Up Scheduling:** Doctors can schedule a follow-up appointment at the end of each consultation, specifying a date range or exact date
- **Automated Reminders:** The system automatically sends reminder notifications to patients via SMS or email before their follow-up date
- **Follow-Up Dashboard:** A dedicated panel in the admin dashboard displays all pending, completed, and missed follow-ups
- **Patient Self-Booking:** Patients receive a special follow-up booking link in their notification that allows them to confirm or reschedule their follow-up appointment within the doctor-specified window
- **Follow-Up Notes:** Doctors can record the purpose and instructions for the follow-up (e.g., “Recheck blood pressure after 2 weeks”)
- **Status Tracking:** Each follow-up has a status — Pending, Completed, Missed, or Rescheduled — for easy monitoring
- **Priority Flag:** High-priority follow-ups (e.g., post-surgery) can be flagged for urgent attention

Table 2: Follow-Up Workflow

Step	Action	Actor
1	Doctor marks follow-up required after consultation	Doctor
2	System creates follow-up record with date & instructions	System
3	Patient receives automated reminder notification	System

4	Patient books or confirms follow-up appointment	Patient
5	Admin views follow-up on dashboard and assigns slot	Admin
6	Doctor reviews previous visit notes before follow-up	Doctor
7	Follow-up marked Complete or Rescheduled after visit	Doctor / Admin

8: Manual Payment Module

The Manual Payment Module is designed for patients who prefer to pay using offline methods such as cash or physical bank transfers. This module ensures that all non-digital payments are properly recorded, verified, and managed by the clinic administration.

8.1 Payment Methods

8.1.1 Cash Payment (At Clinic)

Patients can pay directly at the clinic reception.

Procedure:

1. Patient visits the clinic.
2. Pays the required fee in cash at the reception desk.
3. Receptionist records the payment in the system.
4. System updates payment status to Paid.
5. Printed or digital receipt is provided to the patient.

8.1.2 Manual Bank Deposit (Optional)

Patients can deposit payment directly into the clinic's bank account.

Procedure:

1. Patient visits a bank branch.
2. Deposits the required amount into the clinic's account.
3. Receives a deposit slip or transaction receipt.
4. Submits the proof (photo or reference number) in the system or to admin.
5. Admin verifies the payment manually.
6. Payment status is updated after confirmation.

8.2 Payment Verification Process

- Admin checks submitted proof (receipt, slip, or reference number).
- Confirms payment with bank records (if required).

Marks payment status as:

- **Paid** — if verified
- **Pending** — if under review
- **Rejected** — if invalid proof

8.3 Module Features

8.3.1 Manual Entry by Admin

- Admin/receptionist can enter payment details manually.
- Includes patient ID, amount, date, and method.

8.3.2 Payment Status Tracking

- Each manual payment is linked to an appointment.
- Status is visible to both admin and patient.

8.3.3 Receipt Generation

- System generates a receipt after confirmation.
- Can be printed or shared digitally.

8.3.4 Admin Control

- Only authorized staff can verify or update payments.
- Ensures accuracy and prevents fraud.

8.3.5 Record Maintenance

- All manual payments are stored in the database.
- Useful for audit and reporting purposes.

Table 3: Manual payment method

Payment Method	Location	Verification Method	Handled By
Cash Payment	Clinic	Receipt Confirmation	Receptionist/Admin
Bank Deposit	Bank Branch	Deposit Slip / Reference	Admin

9. Tools and Technologies

To develop the Healthcare Booking System, the following tools and technologies will be used:

9.1 Frontend (User Interface)

- HTML, CSS, JavaScript – To design and create responsive web pages for both patients and clinic staff

9.2 Backend (Server-Side Development)

- Node.js – To handle server-side logic, process user requests, and manage data securely
- Django / Python (alternative) – Optional backend framework if using Python for easier integration with AI modules

9.3 Database

- MySQL – To store user accounts, appointments, and queue information with cloud-based storage and real-time database features
- 9.4 Notifications
- Email API (like SendGrid or Gmail API) – To notify patients about appointments
- Push Notifications [6] – Optional for mobile app version to alert patients in real time

9.6 Optional AI / Analytics

- Python libraries (NumPy, Pandas, scikit-learn) – For simple waiting time estimation or queue prediction
- AI is basic to keep the system practical and easy to implement

9.7 Development and Collaboration Tools

- VS Code / PyCharm / Android Studio – For coding and app development
- GitHub / Git – For version control and collaboration

9.8 Browser and Devices

- The system will work on modern web browsers like Chrome, Firefox, or Edge

10. Flowchart

The flowchart represents the working process of the Healthcare Booking System. It starts when a patient selects a doctor and books an appointment. The system checks whether the selected time slot is available. If it is not available, the process ends with a notification. If available, the appointment is confirmed, and a queue token is generated.

The system stores the details, estimates waiting time, and sends a notification to the patient. The patient can track their position in the queue, while the admin manages and

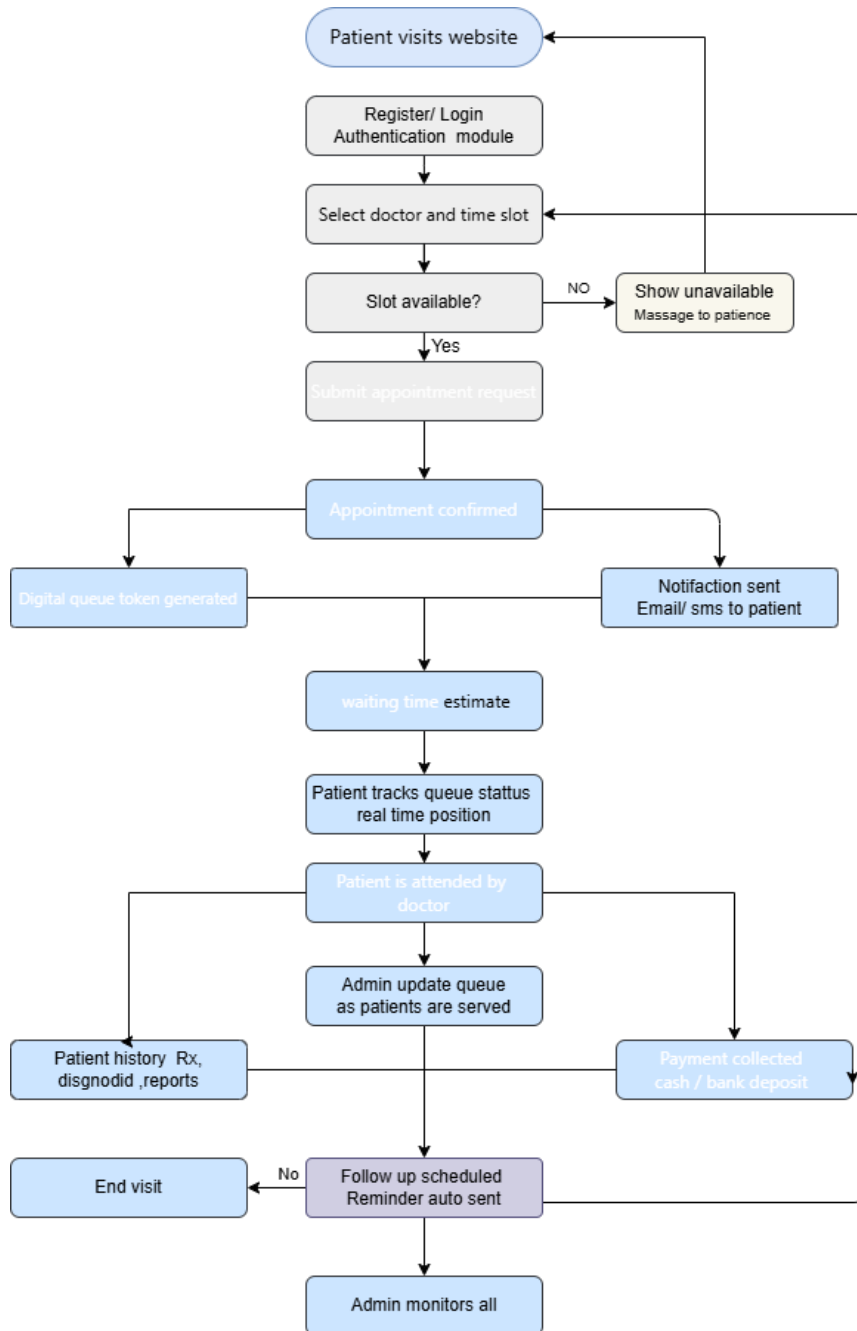


Figure 1: Flow chart

11. Proposed Solution and Anticipated Results

11.1 Proposed Solution

The Healthcare Booking System will provide a practical and easy-to-use web platform for patients and clinic staff [2]. The system allows patients to book appointments, receive digital queue tokens, track their position in real time, access their complete patient history, schedule follow-up appointments, and pay consultation fees securely

online. Clinic staff can manage appointments, monitor queues, and send notifications to patients. Key features of the system include:

- **Online Appointment Booking:** Patients can select a doctor, pick a time slot, and confirm appointments online
- **Digital Queue Management:** Patients receive a queue number and can see their current position in the queue
- **Waiting Time Estimation:** The system predicts waiting time based on queue length and average consultation time [3]
- **Notifications:** Patients get alerts about appointment confirmation, queue updates, follow-up reminders, and payment receipts
- **Admin Dashboard:** Clinic staff can view and manage all appointments, patient queues, doctor schedules, payment records, and follow-ups
- **Patient History:** Full medical history including visits, prescriptions, and lab reports is maintained per patient
- **Follow-Up Management:** Doctors can schedule and track follow-up appointments with automated reminders
- **Online Payments:** Secure payment via Jazz Cash, Easypaisa, or bank transfer with digital receipts

11.2 Anticipated Results

- **Improved Operational Efficiency:** Appointment scheduling and queue management become faster and more organized, reducing manual workload for staff [2]
- **Enhanced Patient Experience:** Patients can book appointments online, track their position in the queue, and receive notifications, which reduces waiting time and frustration [1]
- **Clinic Management:** Staff can manage appointments efficiently, monitor queues, and respond quickly to changes
- **Time-Saving:** Both patients and clinic staff save time by reducing manual booking, waiting, and communication
- **Accessibility:** The system can be accessed via web browsers and optionally on mobile devices for convenience
- **Scalability:** The system can be expanded in the future to include multiple clinics, multiple doctors, or AI-based advanced queue predictions [3]

12. Benefits to Clinics

- Reduces overcrowding and waiting times
- Ensures a smooth workflow for doctors and staff
- Provides secure, trackable digital payments through popular Pakistani platforms
- Maintains complete patient records for better clinical decisions
- Automates follow-up reminders to improve patient retention
- Provides a modern, technology-driven solution suitable for small and medium-sized clinics

13. Schedule of Activities

The development of the Web-Based Healthcare Hospital Management System will follow a structured timeline to ensure all phases are completed efficiently and systematically [4]:

Table 4: Schedule of Activities

Activity	Tentative Time
Requirement Analysis	Week 1–2
System Design & Architecture	Week 3–4
Database Design & Implementation	Week 5–6
Frontend Development	Week 7–8
Backend Development	Week 9–10
Patient History & Follow-Up Module	Week 11–12
Waiting Time & Notification Module Integration	Week 12–13
Online Payment Module Integration (Jazz Cash, Easypaisa, Bank)	Week 13–14
Testing & Debugging	Week 15–16
Documentation	Week 17
Final Review & Submission	Week 18

14. Explanation of Gantt Chart Activities

Requirement Analysis (Week 1–2)

In this phase, system requirements are gathered and finalized [4]. Functional needs such as appointment booking, queue management, patient history, follow-up scheduling, notifications, and online payment integration are identified. This step ensures a clear understanding of the project scope before development begins.

System Design & Architecture (Week 3–4)

The overall system structure is designed during this phase [4]. This includes database schema design, system flow diagrams, payment flow diagrams, and user interface wireframes. The aim is to create a clear blueprint for implementation.

Database Design & Implementation (Week 5–6)

In this phase, the database is designed and implemented to store appointments, queue tokens, patient history, follow-up records, and payment transactions. Firebase or MySQL is used to ensure reliable data storage and real-time access [5].

Frontend Development (Week 7–8)

The user interface is developed using HTML, CSS, and JavaScript. This phase focuses on creating responsive pages for patients and clinic staff, including the payment interface and patient history dashboard, ensuring usability across different devices.

Backend Development (Week 9–10)

Server-side logic is implemented to handle appointment requests, queue updates, patient record management, follow-up scheduling, and communication between the frontend and database. This phase ensures secure and efficient system operations [4].

Patient History & Follow-Up Module (Week 11–12)

The patient history module is developed to record and retrieve full medical records per patient. The follow-up scheduling module is integrated to allow doctors to assign follow-up dates and for patients to receive automated reminders.

Waiting Time & Notification Module Integration (Week 12–13)

Basic rule-based or AI-assisted logic is integrated to estimate waiting time [3]. Notification services such as email or SMS are also implemented to inform patients about appointments, queue updates, and follow-up reminders [6].

Online Payment Module Integration (Week 13–14)

The Jazz Cash and Easypaisa payment APIs are integrated for mobile wallet payments. IBFT bank transfer support is also configured. Invoice generation, payment status tracking, and digital receipt delivery are implemented and tested.

Testing & Debugging (Week 15–16)

The complete system is tested to identify and fix errors [4]. Functional testing, integration testing, payment gateway testing, and usability testing are performed to ensure system reliability.

Documentation (Week 17)

Project documentation is prepared, including system design details, user manuals, API integration guides, and technical explanations. This phase ensures clarity for future maintenance and evaluation.

Final Review & Submission (Week 18)

The system is reviewed for final improvements. The project is prepared for presentation, demonstration, and official submission.

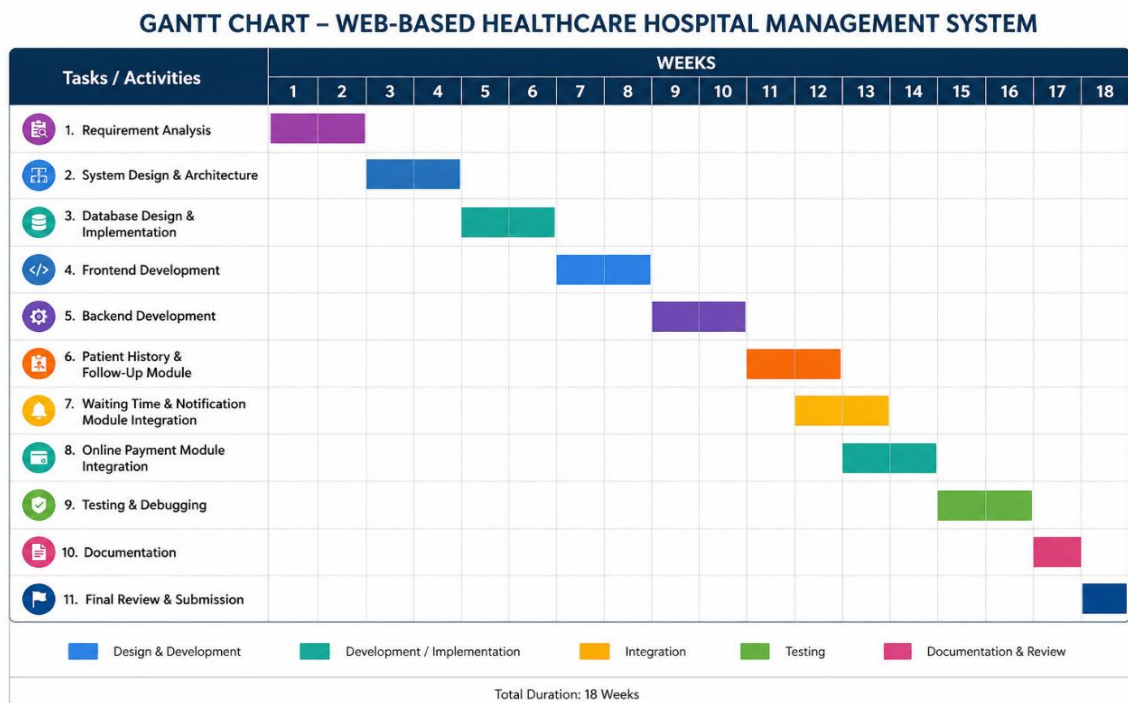


Figure 2: Gantt Chart

15. Future Enhancements

Future enhancements may include a mobile application version, integration with electronic health record systems, advanced machine learning models for appointment optimization, and multilingual support for broader accessibility. These enhancements can further improve system usability and commercial potential [3].

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